



**Faculty of Electrical Engineering
Department of Measurement**

Bachelor's Thesis

LionKey (security key) NFC support

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<https://github.com/Matesin/bachelor-thesis>

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Guidelines:

For the LionKey project: An open-source FIDO2 USB Security Key, implement basic support for NFC technology.

The work consists of the following parts:

- 1) Familiarization with the existing codebase and the technologies needed to implement NFC.
- 2) Implement basic NFC functionality on the STEVAL-ST25R3916B hardware.
- 3) Demonstrate basic functionality - e.g., authentication on <https://webauthn.io>

Bibliography / sources:

<https://github.com/pokusew/lionkey>

https://dspace.cvut.cz/bitstream/handle/10467/123886/F3-DP-2025-Endler-Martin-FIDO2_USB_Security_Key.pdf?sequence=1&isAllowed=

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Acknowledgement

Abstrakt / Abstract

Tato bakalářská práce se zabývá implementací podpory technologie NFC pro hardwarový bezpečnostní klíč LionKey, který slouží jako externí FIDO2 autentizátor. Cílem práce je rozšířit existující zařízení o komunikační rozhraní umožňující jeho použití také se zařízeními vybavenými NFC čtečkou, zejména pak s mobilními telefony.

Práce popisuje teoretické principy technologie NFC, související komunikační standardy a jejich vztah k použití bezpečnostních klíčů v autentizačních mechanismech. Dále se věnuje návrhu a realizaci NFC rozhraní pro platformu LionKey, integraci do existující architektury klíče a ověření funkčnosti výsledného řešení.

Výsledkem práce je funkční implementace NFC komunikace pro klíč LionKey s využitím platformy STEVAL-D25R16B. Výsledné řešení je veřejně dostupné na GitHubu a rozšiřuje možnosti použití zařízení na stránkách podporujících WebAuthn.

Klíčová slova: NFC, WebAuthn, CTAP2, bezpečnostní klíč

Překlad titulu: LionKey (bezpečnostní klíč) rozšíření NFC

This bachelor's thesis focuses on implementing NFC support for the LionKey hardware security key, which serves as an external FIDO2 external authenticator. The aim of this thesis is to extend the capabilities of the existing device by implementing an NFC communication interface on top of the existing solution, allowing the key to communicate with devices equipped with an NFC reader, especially mobile phones.

The thesis describes the theoretical principles of NFC technology, the related communication standards, and their relevance to the use of security keys in authentication mechanisms. It also focuses on the design and implementation of an NFC interface for the LionKey platform, its integration into the existing device architecture, and the verification of the resulting solution.

The result of this thesis is a functional implementation of NFC communication for the LionKey security key using the STEVAL-D25R16B platform. The resulting implementation is publicly available on GitHub. It extends the device's capabilities by allowing its use with services supporting WebAuthn.

Keywords: NFC, WebAuthn, CTAP2, security key

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Chapter 1

Introduction

Passkeys have become a **popular means of authentication** on the Internet. Along with more conventional means of passkeys, users are also adopting **hardware security keys** as a **safer way to store their credentials**. Hardware security keys may provide an additional layer of security compared to platform authenticators. They offer stronger isolation of private keys and ensure their non-exportability, thereby reducing the risk of credential compromise even if the host device is not trusted.

One such project is the **LionKey**, an open-source hardware security key developed at **CTU FEE** by **Ing. Martin Endler**. The device is compliant with FIDO2 standards; however, it currently **supports only CTAP over USB** (CTAPHID). This means that the key cannot be used with devices that do not support USB-based communication, notably mobile phones.

1.1 Thesis Objective

The objective of this thesis is to **extend** the **communication capabilities** of the LionKey hardware security key by **implementing an NFC-based communication** interface on top of the existing CTAPHID solution, in order to further improve user experience when using the authenticator.

The NFC interface is realized using the STEVAL-D25R3916B board, which is interfaced with the ST Nucleo-H533RE microcontroller in accordance with the original specifications of the LionKey project.

1.2 Thesis Structure

The thesis is organized into six main parts:

The **Introduction** outlines the context of the work and formulates the objectives of the thesis.

Chapter 2 presents the theoretical background of passkeys and motivates their use in modern authentication systems.

Chapter 3 introduces the fundamental principles of NFC technology, focusing on aspects relevant to the proposed solution.

Chapter 4 details the design, hardware used, and implementation of the proposed solution.

Chapter 5 describes the results achieved by this implementation and its applicability to practice.

Chapter 6 summarizes the results achieved and discusses the general outcome of the thesis.

Chapter 2

An Alternative Authentication Method

This chapter defines the motivation for the use of an alternative authentication method.

The Internet has become an integral part of daily life for most of the world's population. In December 2025, approximately **6.047 billion people** (73.2% of the global population) were **Internet users**, representing a 14% increase compared to 2020 [1]. This growth, along with an increasing reliance on online services, has been accompanied by an increase in cyberattacks and personal data breaches.

This development highlights a fundamental weakness of password-based authentication. According to the Verizon Data Breach Investigations Report, approximately **88% of breaches** in 2025 involved **stolen credentials** [2]. Passwords are therefore inherently vulnerable to theft, reuse, and brute-force attacks.

In response, organizations have **increasingly adopted multi-factor authentication (MFA)**, which supplements passwords with additional verification factors such as one-time codes, authentication applications, or trusted devices. The adoption of MFA has grown significantly; for example, **83% of organizations required MFA** for all access to IT in 2024 [3]. Despite variations across surveys, it is clearly observable that **MFA has become a popular safety measure**.

2.1 Passkeys

Despite the growing usage of MFA, Ing. Martin Endler argues in his diploma thesis that many **existing authentication methods** “**do not provide sufficient security** and hinder the user experience” [4]. Passkeys – standardized by the FIDO Alliance – have emerged as a response to these limitations.

The FIDO Alliance **defines a passkey** as “...[an] authentication **credential** based on FIDO standards that allows a user to **log into apps and websites** with the same process they use to unlock their device (biometrics, PIN or pattern). Passkeys rely on **asymmetric cryptography**, where the private key is securely stored on the user's authenticator, while the corresponding public key is registered with the service ” ¹.

According to data published by the FIDO Alliance, passkey support has already been **adopted** by approximately a **fifth** of the world's **top 100 websites** and services [5]. Furthermore, major technology providers, including Apple, Google, and Microsoft, have integrated passkey support into their platforms, allowing further support for this authentication method.

¹ <https://fidoalliance.org/passkeys/>

■ 2.1.1 Hardware security keys

Although passkeys are often implemented as platform authenticators integrated into user devices, the **FIDO2** framework also supports **external authenticators** in the form of **hardware security keys**. These devices **store cryptographic credentials** on dedicated hardware and **perform authentication independently** of the host system and only **upon user interaction**, providing an additional layer of security.

Currently, the hardware security key market is dominated by several major vendors, most notably **Yubico** with its Security Key Series. Other notable solutions include the Google Titan Security Key developed by Google and the SecurID authenticators made by RSA Security. In addition, several smaller or less widely cited manufacturers provide compatible FIDO2 security keys, including FEITIAN Technologies, Swissbit, and TrustKey.

Quantitative estimates of market share are difficult to obtain, as publicly available data are limited and often based on proprietary methodologies. **Estimates suggest**, as of April 2026 [6] that **Yubico accounts for approximately 32% and RSA Security for 24%** of the two-factor authentication **solutions**. However, these figures reflect only the customer base observed by the source platform and should be interpreted as indicative rather than representative of the global market share of hardware security keys.

A distinguishing **characteristic** of most commercially available hardware security keys is their **closed-source firmware**. Open-source alternatives do exist but remain rare: the first publicly known example was SoloKeys (developed 2018–2024, now discontinued), followed by OpenSK by Google and Nitrokey. The **presented work** contributes to this space by providing the **first open-source implementation of CTAP2 over NFC** in a security key **written in C**.

■ 2.2 The FIDO2 Framework

The **FIDO2** framework, developed by the FIDO Alliance in collaboration with the W3C, is the current **industry standard** for **strong, phishing-resistant and passwordless authentication**. It is an open framework that uses public-key cryptography.

FIDO2 is composed of two complementary core specifications:

- **WebAuthn (Web Authentication)**: A W3C standard API that enables web applications to create and use public-key credentials. It defines how a browser or platform (the “Client”) communicates with a web service (the “Relying Party”) to register and authenticate users.
- **CTAP (Client-to-Authenticator Protocol)**: A FIDO Alliance specification that defines how a Client communicates with an authenticator (such as a hardware security key or a smartphone) to fulfill WebAuthn requests.

The CTAP standard defines two primary versions:

- **CTAP1 (formerly U2F)**: Supports legacy FIDO U2F credentials.
- **CTAP2**: The current standard for FIDO2, which improves user verification methods, most notably by implementing a PIN/User Verification (UV) authentication model.

A **detailed breakdown** of this framework, including a detailed description of the CTAP protocol, can be **found** in the **work of Endler** [4], along with a more comprehensive insight into the problematic of hardware security keys, including the cryptographic mechanisms that are commonly used.

■ 2.2.1 Post-Quantum Relevance of the CTAP Protocol

To further extend Endler’s overview, we would like to introduce the reader to the implications of quantum computing on the CTAP protocol.

The version of CTAP implemented in this project (CTAP 2.1) does not provide native post-quantum (PQ) security, as it relies on conventional public-key cryptographic primitives that could be compromised by sufficiently capable quantum adversaries. Nevertheless, the current protocol architecture is considered to be PQ ready. In particular, its PIN/UV authentication model requires active user verification and physical presence, while the protocol itself can be adapted to stronger cryptographic primitives without a fundamental redesign. Therefore, a full quantum-resistant operation could be achieved by replacing the current asymmetric mechanisms with post-quantum key encapsulation mechanisms and digital signature schemes, provided that such changes are introduced consistently across both CTAP and WebAuthn within the FIDO2 framework [7].

Chapter 3

NFC

This chapter introduces the principles of the **Near Field Communication (NFC)** technology. Due to the extensive nature of NFC standardization, the **main focus** is placed on those of its **mechanisms** that are **relevant** to the presented **implementation**.

Other technologies within the NFC ecosystem are discussed only briefly to provide context and completeness.

3.1 Introduction to NFC

Near Field Communication (NFC) is a short-range wireless communication technology derived from radio-frequency identification (RFID) systems. It enables **bidirectional data exchange** between devices in close proximity, typically within a range of up to 10 cm, with reliable operation occurring at distances of approximately 4 cm [8].

NFC operates at a **carrier frequency of 13.56 MHz** and is based on **inductive coupling** between **loop antennas**. In this mechanism, an active device generates an electromagnetic field, while a passive device modulates this field to transmit data.

The NFC technology is standardized by international organizations, primarily by joint standards of the International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC), abbreviated as ISO/IEC, and the NFC Forum¹. The key specifications include ISO/IEC 14443, which defines proximity card communication (used in NFC-A and NFC-B), and ISO/IEC 18092, which specifies peer-to-peer communication. In addition, the NFC Forum defines higher-level protocols and data formats to ensure interoperability across devices.

NFC supports three primary operating modes: reader/writer mode, card emulation mode, and peer-to-peer mode. From the perspective of this thesis, card emulation mode is of particular importance, as it allows the device to behave as a contactless smart card and participate in ISO-DEP communication with an external reader.

3.2 Device Types

NFC devices can be broadly categorized into **active** and **passive** types.

Active devices, often referred to as **universal NFC frontends**, are equipped with their **own power source** and are capable of **generating an electromagnetic field**. This enables them to initiate communication and control the timing and parameters of the exchange.

¹ <https://nfc-forum.org>

In contrast, **passive devices** (commonly implemented as **dynamic** NFC tags) do **not require an internal power supply**. Instead, they rely on the energy harvested from the electromagnetic field generated by an active device (**Energy Harvesting**). **Communication** from the passive side is achieved through **load modulation**, i.e., by varying the impedance of the antenna circuit, which in turn modulates the carrier field.

3.3 Network Hierarchy

In NFC communication, two roles are distinguished according to which device initiates the interaction: the **initiator** and the **target**. These roles are defined at the protocol level and are independent of the specific hardware capabilities of the devices.

■ Initiator

The device that actively starts the communication and generates the initial RF field. It is responsible for polling and controlling the communication sequence, as is typically the case for an NFC reader.

■ Target

The device that responds to the initiator. Depending on the communication mode, it can passively modulate the initiator's field or actively generate its own signal. Typical examples include contactless smart cards, NFC tags, or peer devices.

It is important to distinguish these roles from the physical ability to generate an electromagnetic field. Although the initiator is typically the field-generating device, this is not strictly required in all communication modes.

NFC defines two communication modes based on how the electromagnetic field is generated and how the devices are powered: **active mode** and **passive mode**.

■ Active Mode

In active mode, both the initiator and the target are equipped with their own power sources and are capable of generating an RF field. Communication is performed through alternating transmission, where each device generates its own carrier field during its transmission phase. This mode is used primarily in peer-to-peer communication scenarios.

■ Passive Mode

In passive mode, only the initiator generates the RF field. The target does not have its own power source and utilizes energy harvesting. Data transmission from the target is achieved through load modulation of the existing carrier field. This mode is characteristic of typical NFC-A and NFC-B interactions, such as communication with contactless smart cards and NFC tags.

3.4 Operating Modes

The NFC Forum defines **three** primary **operating modes** that determine the role and behavior of a device within a communication session.

■ Reader/Writer Mode

In the reader/writer mode, the NFC device operates as an initiator and interacts with passive NFC tags. The device generates an electromagnetic field and communicates with tags compliant with standards such as ISO/IEC 14443 (Type 2 and Type 4

Tags) or ISO/IEC 15693 (Type 5 Tags). This mode is commonly used for applications such as reading NDEF records, accessing contactless cards, or configuring embedded devices.

■ **Card Emulation Mode**

In card emulation (CE) mode, the NFC device behaves as a contactless smart card and acts as a target. It responds to commands issued by an external reader, typically using ISO/IEC 14443 protocols. This mode is widely used in applications such as contactless payments, access control, and mobile wallets. This mode has two variants of implementation. It can either be implemented on a general-purpose MCU or using a dedicated secure element (SE) hardware, depending on the use case. The majority of security keys supporting NFC uses the SE variant (as implied, for example, by the Yubikey tech manual [9] or by the Google Titan documentation [10]).

■ **Peer-to-Peer Mode**

The peer-to-peer mode enables half-duplex communication between two NFC-enabled devices, where both devices can alternately assume the roles of initiator and target. Communication in this mode is defined by ISO/IEC 18092 and the NFC Forum Logical Link Control Protocol (LLCP). It is primarily used for data exchange scenarios such as device pairing or file transfer, although its practical use has declined in favor of higher-throughput wireless technologies.

These operating modes build upon the previously described concepts of initiator and target roles, as well as active and passive communication modes. Their combination defines the functional behavior of NFC devices in real-world applications.

■ **3.5 Types of NFC Tags**

■ **3.5.1 NFC-A**

NFC-A is based on the ISO/IEC 14443 Type A specification. Data transmission in this standard employs Miller (delay) encoding in combination with Amplitude Shift Keying (ASK) modulation, with a typical data rate of 106 kbit/s.

Binary information is conveyed by variations in the amplitude of the carrier signal. Specifically, logical values are not represented by absolute voltage levels, but rather by the presence or absence of modulation and by transitions between minimum and maximum signal amplitude. In Miller encoding, these transitions are time-dependent, meaning that the information is encoded both in the timing and structure of the signal changes, which improves robustness against noise and enables reliable synchronization between communicating devices.

■ **3.5.2 NFC-B**

NFC-B is based on the ISO/IEC 14443 Type B specification. It uses NRZ-L (Non-Return-to-Zero Level) encoding for data representation, combined with Binary Phase Shift Keying (BPSK) for subcarrier modulation during card-to-reader transmission. For reader-to-card transmission, a 10% ASK modulation is applied directly to the carrier, where a reduction in carrier amplitude to approximately 90% of its nominal level represents a logical 0, while the full amplitude represents a logical 1.

Due to the lower modulation index compared to NFC-A, the carrier signal remains more stable, resulting in improved robustness against electromagnetic interference. NFC-B

is used primarily in electronic identification documents such as electronic passports, where enhanced reliability and support for more complex security mechanisms are required [11].

3.5.3 NFC-F

NFC-F is based on the JIS X 6319-4 specification and was originally developed by Sony under the commercial name FeliCa. Unlike NFC-A and NFC-B, NFC-F is not strictly derived from the ISO/IEC 14443 family, but instead follows a different protocol structure defined in Japanese industrial standards.

A key distinguishing feature of NFC-F is its higher data rates, typically 212 kbit/s or 424 kbit/s, which makes it suitable for latency-sensitive applications. The technology is used primarily in Japan, particularly in transportation systems, contactless payment infrastructures, and access control solutions.

However, due to its regional focus and protocol differences, NFC-F is not universally supported by all NFC readers. Consequently, its applicability in general-purpose NFC implementations is limited.

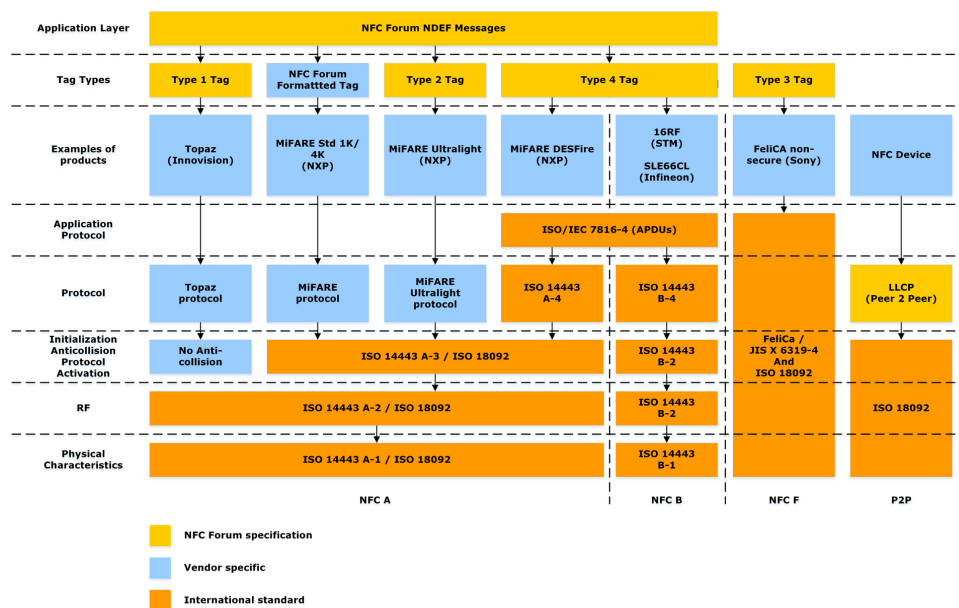


Figure 3.1. NFC Protocol Stack [12]

3.6 NFC Communication Protocols

NFC communication is built upon a layered protocol stack that defines how data is exchanged between devices after the initial activation phase. At the transport and higher communication layers, two primary protocols are used: ISO-DEP and NFC-DEP. In addition, several auxiliary protocol layers defined by the NFC Forum provide standardized data exchange formats.

3.6.1 ISO-DEP

The **ISO-DEP** (ISO Data Exchange Protocol), specified in ISO/IEC 14443-4 [13], defines a transport layer for reliable data exchange between an NFC initiator and a proximity

integrated circuit card. It extends the lower NFC-A/B protocols by introducing a block-oriented, half-duplex communication mechanism that enables structured and robust transmission of application data.

3.6.1.1 Activation and Parameter Negotiation

Before data exchange can begin, ISO-DEP requires an activation phase in which the initiator selects the target and negotiates communication parameters. This is performed using the RATS (Request for Answer To Select) and ATS (Answer To Select) mechanism.

First, the reader sends a RATS command. The RATS consists of a command byte and a parameter byte, where the upper nibble encodes the Frame Size for the Device Input (FSDI), defining the maximum frame size supported by the reader. The lower nibble may contain an optional Card Identifier (CID), which is used for logical addressing during communication.

The ATS response consists of a total length value (TL), format byte (T0), optional interface bytes (TA, TB, TC) and optional Historical Bytes (up to 15 bytes). These fields define key communication parameters, including timing constraints and frame sizes.

TL	T0	TA(1)	TB(1)	TC(1)	T1	T_k
		interface				historical bytes

Table 3.1. ATS structure.

The interface bytes can contain these values:

- **Frame Waiting Time (FWT)**

Specifies the maximum time the initiator waits for a response from the target. It is derived from the Frame Waiting Time Integer (FWI) and directly influences the allowable processing time of the card. Violating this constraint typically results in termination of the NFC session.

- **Waiting Time Extension (WTX)**

Allows the target to request additional processing time beyond the configured FWT. This mechanism is essential for operations involving computationally intensive tasks, such as cryptographic processing.

- **Frame Size (FSC/FSD)**

Defines the maximum payload size of a single frame for the card (FSC) and the device (FSD), determining whether chaining is required.

- **Frame Guard Time (FGT)**

Specifies the minimum delay between consecutive frames, ensuring proper synchronization between devices.

- **Device Identifier (DID)**

Allows logical addressing of devices in more complex communication scenarios.

- **Historical Bytes (HB)**

Optional data included in the ATS, providing additional information about the capabilities of the card.

Correct configuration and handling of these parameters is essential to ensure compatibility with readers and mobile devices. Strict adherence to timing constraints, such as the frame waiting time, is required to prevent premature session termination.

3.6.1.2 Data Exchange

Communication in **ISO-DEP** is organized into protocol data units (blocks), which are exchanged between the initiator and the target. Three types of blocks are defined: **I-Blocks** to carry application data, **R-Blocks** for acknowledgments and retransmission requests, and **S-Blocks** for control operations such as waiting time extension or protocol deactivation. This structure ensures reliable communication even in the presence of transmission errors.

3.6.1.3 APDU

ISO-DEP serves as the transport layer for **Application Protocol Data Units (APDUs)**, which implement a command-response communication model. APDUs are a standardized data format used in smart card systems for structured information exchange defined by ISO/IEC 7816-4.

In this model, the initiator sends a **command APDU (C-APDU)**, and the target responds with a **response APDU (R-APDU)**. This abstraction allows higher-level protocols, including smart card applications and CTAP over NFC, to operate independently of the underlying transmission mechanism while relying on ISO-DEP for reliable delivery.

A command APDU (C-APDU) consists of a header (CLA, INS, P1, P2) and an optional payload defined by the fields Lc (length of command data) and Le (expected length of response data). The response APDU (R-APDU) may contain response data and is always terminated by a two-byte status word (SW1, SW2), indicating the result of the operation.

APDUs can be encoded in short or extended format. In the short format, the length fields are limited to one byte, allowing payloads of up to 255 bytes. The extended format uses two-byte length fields, enabling payloads of up to 65,535 bytes.

In the short format, the C-APDU consists of the mentioned 4-byte header followed by the optional fields Lc, Data (0–255 bytes) and Le, where both Lc and Le are encoded using a single byte.

CLA	INS	P1	P2	Lc	Data	Le
1 byte	1 byte	1 byte	1 byte	0/1	0–255	0/1

Table 3.2. Byte-level structure of short C-APDUs.

In the extended format, the presence of the leading 0x00 byte indicates extended length encoding, followed by two-byte length fields for Lc and Le.

Response APDUs consist of an optional data field followed by a mandatory two-byte status word (SW1, SW2).

CLA	INS	P1	P2	00	Lc(Hi)	Lc(Lo)	Data/Le
1B	1B	1B	1B	ext. marker	length		variable

Table 3.3. Byte-level structure of extended C-APDUs.

Data	SW1	SW2
(optional) response data	status	word

Table 3.4. General structure of a R-APDU.

```

CE: start waiting for command
CE: negotiated max frame size: 252
CE: start RX successful
Received APDU (13 bytes): 00A4040008A0000006472F0001
NFC: APDU parsed successfully
NFC: FIDO AID selected
Sent APDU (10 bytes): 4649444F5F325F309000

```

Figure 3.2. Device log of processing an APDU and selecting the FIDO applet.

The practical structure of APDU communication can be illustrated on the **following example**:

C-APDU:	00	A4	04	00	08	A0	00	00	06	47	2F	00	01
						_____ Data _____/							
	CLA	INS	P1	P2	Lc								
R-APDU:	46	49	44	4F	5F	32	5F	30	90	00			
	_____ Data _____/								\SW1 SW2/				

In this example, the C-APDU consists of the following fields: CLA = 0x00 (interindustry class byte - a standard entry point for commands), INS = 0xA4 (SELECT instruction), P1 = 0x04 (selection by name) and P2 = 0x00. The field Lc = 0x08 specifies the length of the data field, which contains the 8-byte FIDO application identifier (AID) (A0 00 00 06 47 2F 00 01). Upon processing this message, the device responds with an R-APDU.

The R-APDU contains the response data followed by the status word. The data field corresponds to the ASCII string “FIDO_2_0”, while the status word 0x9000 indicates successful execution of the command.

3.6.1.4 Chaining mechanisms

Within ISO-DEP, APDUs are transported using I-Blocks. Since the maximum size of a single ISO-DEP frame is limited by the negotiated frame size, APDUs exceeding this size must be fragmented. This is achieved through **ISO-DEP chaining**, where a single APDU is split across multiple I-Blocks and reassembled by the receiving side.

In addition to transport-level fragmentation, large responses may also be handled at the application level. In the case of short APDUs, the response length is limited by the value of the Le field. If the response exceeds this limit, it can be delivered in multiple

steps using the status word mechanism (e.g., $SW1 = 0x61$), where the client issues the GET RESPONSE command to retrieve the remaining data.

The two chaining mechanisms operate at different layers: ISO-DEP chaining ensures reliable transmission of a single APDU over multiple frames, while application-level response handling enables the transfer of data exceeding the limits of the APDU format itself.

■ 3.6.2 NFC-DEP

The **NFC-DEP** (NFC Data Exchange Protocol), defined in ISO/IEC 18092, is designed for peer-to-peer communication between two **active** NFC devices. Unlike ISO-DEP, which follows a reader–card model, NFC-DEP enables symmetric communication, allowing both devices to alternate between initiator and target roles.

In contrast to ISO-DEP, **NFC-DEP does not utilize APDUs** and does not follow a strict command–response model. Instead, communication is based on **message-oriented data exchange**, making it more suitable for peer-to-peer scenarios rather than smart card emulation.

For this reason, NFC-DEP is **not applicable** to the desired goal of this thesis, as FIDO2 authentication over NFC relies on APDU-based communication provided by ISO-DEP.

Chapter 4

Implementation

In this chapter, the **technologies used** in the implementation of NFC for the LionKey security key as well as the **software implementation** are described. The reasons for system design decisions come mainly from facts stated in Chapter 3.

4.1 NFC technologies used

After evaluating the available NFC operating modes and specifications, it becomes evident that the Card Emulation Mode is the only mode compatible with the intended use case. The objective is for the device to behave as a passive entity, analogous to a contactless smart card, rather than as an active initiator.

In the ideal case, card emulation based on Secure Element (SE) would be used. However, this is not feasible with respect to the existing LionKey hardware architecture. This approach requires the presence of a dedicated SE capable of isolated execution and secure key storage, which is not available in the current design. Consequently, the implementation must be based on host-based card emulation, where all protocol handling is performed by the microcontroller (MCU).

The required functionality is achieved most appropriately through the emulation of a Type 4 NFC Forum Tag. In contrast to simpler tag types, Type 4 Tag operation is based on the ISO-DEP protocol and supports structured command–response communication via APDUs. This enables the implementation of a well-defined state machine and facilitates higher-level protocols, which are essential for authentication mechanisms such as FIDO2.

Taking these constraints into account, the implementation adopts NFC-A as the underlying transmission technology. NFC-A provides the broadest interoperability across existing NFC readers, particularly in mobile devices, and is the de facto standard for ISO-DEP–based communication and for Type 4 tag implementations.

Our implementation follows the FIDO CTAP2 standard [14].

4.2 Hardware

The hardware chosen for the implementation is the STEVAL-D25R3916B, a daughter board of the STEVAL-25R3916B platform manufactured by STMicroelectronics [15]. This kit is an evaluation board for the 25R3916B NFC chip, which turned out to be the best option for the purpose of this thesis.

The 25R3916B chip belongs to the STMicroelectronics universal NFC device family. It

supports all main NFC operating modes, including reader/writer mode, peer-to-peer communication, and card emulation; however, it does not provide support for secure element-based card emulation.

Unlike dedicated NFC tag ICs with fixed memory structures, the chip operates as a low-level RF frontend, exposing frame-level control to the application layer. Thanks to this, it enables software-driven card emulation, allowing the host microcontroller to implement the protocol logic.

4.2.1 Board Specifications

The STEVAL-D25R3916B embeds the **ST25R3916 chip** with an antenna etched on the PCB and the related Very High Bit Rate (VHBR) tuning circuit, allowing it to communicate at higher speeds.

In terms of host communication, the board provides both the Serial Peripheral Interface (SPI) and Inter-Integrated Circuit (I2C) interfaces, complemented by an interrupt request line (IRQ) used to signal events to the MCU.

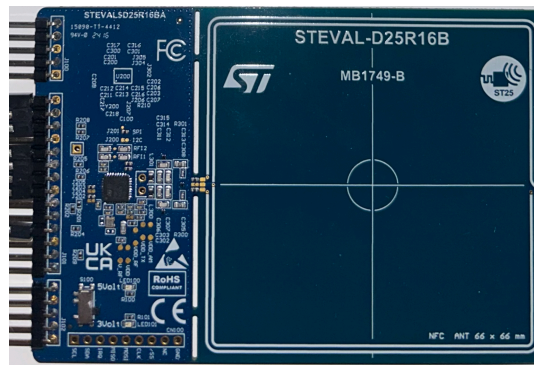


Figure 4.1. The STEVAL-D25R3916B board.

4.2.2 Note: Energy Harvesting

A significant limitation of the selected device is its absence of energy harvesting capabilities. Consequently, the security key cannot operate as a fully passive, standalone authenticator and requires an external power source during operation. It is important to note that achieving true standalone functionality would require not only an NFC frontend capable of efficient energy harvesting but also a substantially lower-power MCU.

The power demands of the current hardware platform exceed what can be reliably supplied by the electromagnetic field generated by typical NFC readers, making passive operation impossible under these conditions.

4.3 Middleware

The middleware used in this project is the STSW-ST25RFAL002 package, provided by STMicroelectronics as an implementation of the RFAL (RF Abstraction Layer) software for the ST25R3916B NFC frontend. RFAL represents a hardware-oriented abstraction

layer that encapsulates low-level device control and RF protocol handling. It provides a set of primitives for device initialization, RF configuration, register-level access, and data exchange.

In our implementation, RFAL forms the intermediary layer between the LionKey application and the NFC frontend. The application exchanges data and control requests with RFAL, while RFAL is responsible for direct interaction with the NFC device, low-level NFC control, communication parameters negotiation and subsequent ISO-DEP R-APDU fragmentation (i.e. when the size APDU being sent is larger than the negotiated frame size). The higher-level communication logic, including the NFC state machine and protocol flow control including short APDU chaining, is implemented in the application layer.

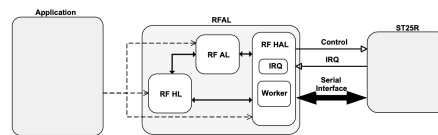


Figure 4.2. Communication flow between the Application (LionKey) and the NFC device using RFAL. (Source: STMicroelectronics STSW-ST25RFAL002 Data Brief)

4.4 Communication with the NFC Frontend

This section describes the communication interface between the MCU and the NFC frontend.

4.4.1 SPI Interface

The communication between the MCU and the NFC frontend is implemented using SPI. This choice is primarily motivated by the strict FWT limits.

Compared to I2C, SPI provides higher data throughput, lower communication latency, and fully deterministic timing due to its synchronous nature and absence of bus arbitration. These properties are critical to ensure protocol compliance. Any excessive delay in data transfer between the front end and the MCU may lead to FWT violations, thus prematurely terminating the NFC session.

In our implementation, SPI operates in a master-slave configuration, with the MCU acting as the master and ST25R3916B acting as the slave device. Chip select (CS) is manually controlled via a GPIO pin to ensure precise control over transaction boundaries.

4.4.2 Interrupt Handling

The IRQ line is asserted by the NFC frontend to indicate events such as reception of RF frames, completion of transmission, state changes in the RF field or error conditions.

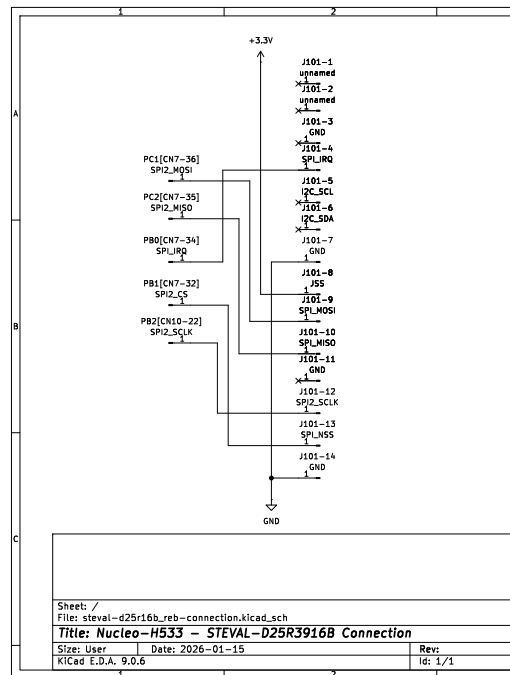


Figure 4.3. KiCad scheme showing how the STEVAL-D25R3916B board is connected to the Nucleo-H533 MCU.

Upon receiving the interrupt, the MCU reads the corresponding status registers through SPI and sends the event to RFAL. This mechanism enables prompt processing of incoming ISO-DEP frames and immediate reaction to transmission errors.

4.5 Software Implementation

In terms of code, the software implementation follows the practices set by Ing. Endler in the original project. It is written in C with no dynamic memory allocations utilizing a layered design that separates the NFC transport layer, APDU processing, and higher-level FIDO logic.

The NFC functionality is implemented using a set of state machines integrated into the existing application architecture, with the NFC worker function being ran periodically from the main application loop.

The primary state machine models the operational state of the NFC frontend and distinguishes between the following states:

- Inactive (Idle) state
The device is not within the range of an NFC reader.
- Active state
CE mode is active and data exchange with the reader is in progress.
- Reset state
The communication session has ended and the NFC runtime context is reinitialized to its default configuration.

4.5.1 Transaction Handling

To ensure non-blocking operation and enable concurrent execution of other critical components in parallel with NFC, the card emulation task is implemented as a separate state machine. This secondary state machine governs the progression of individual NFC transactions, including reception and transmission phases.

Upon receiving an APDU from the reader, the message is parsed and processed according to its semantics. The parser supports all defined APDU formats and handles edge cases according to the mentioned ISO specification. The implementation supports both CTAP2-related communication and basic NDEF communication. The NDEF state machine provides the user with an URI record that directs the user to the lionkey.dev website.

Upon receiving, parsing and, in case of a chained APDU request, concatenating a CTAP2 command, this is then handled by the CTAP API provided in the original LionKey solution.

Management of the communication state machine is performed by periodic polling of the exchange status provided with the RFAL library with errors handled in accordance with the ISO specification.

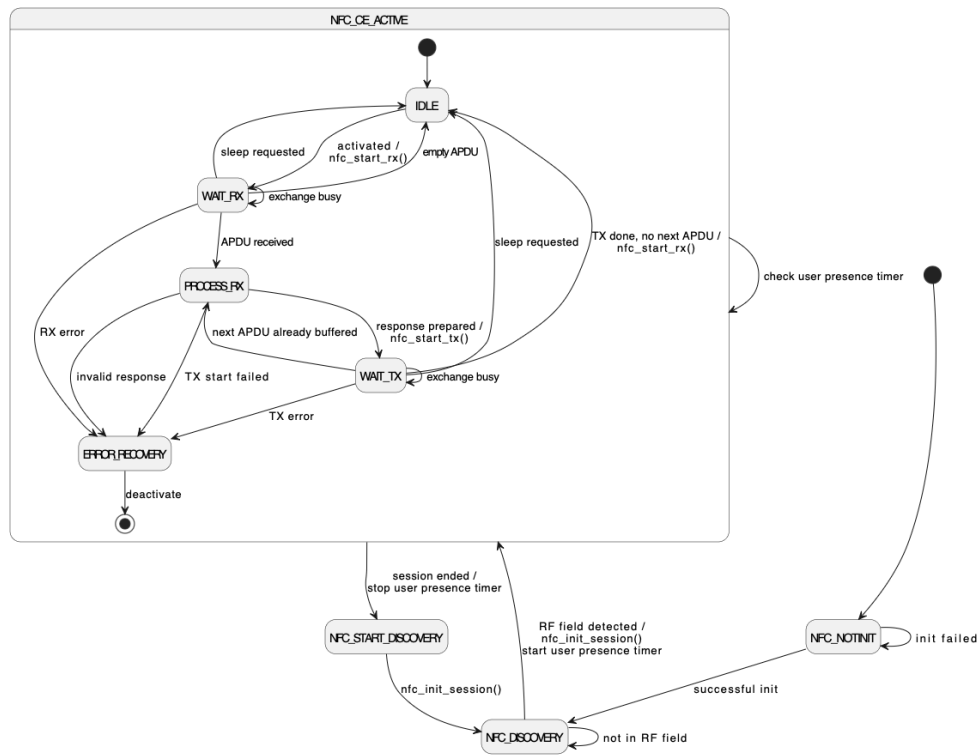


Figure 4.4. State diagram of the implemented state machine.

Chapter 5

Results and Evaluation

This chapter describes how the correctness of the implementation was tested.

5.1 Methodology

The evaluation process was structured in two stages, corresponding to different layers of the NFC communication stack.

First, the ability of the device to correctly respond during the activation phase was verified. This included transmission of fundamental identification and protocol parameters, namely ATQA (SENS_RES), UID, SAK, and ATS. The tests were performed using an external NFC reader under controlled conditions, enabling detailed observation of the anti-collision and activation sequence. Successful completion of this stage confirms correct operation at the physical and protocol initialization layers as defined by ISO/IEC 14443 and the ISO-DEP protocol.

In the second stage, the communication capabilities of the device were evaluated. Both supported application layers, CTAP and NDEF URI handling, were tested. Interoperability testing was conducted using consumer mobile devices as well as personal computers.

The mobile devices used for testing were multiple iPhone devices ranging from iPhone 11 to iPhone 15 with both iOS 18 and iOS 26, and a Samsung Galaxy S22+, representing the dominant mobile platforms, iOS and Android. Regarding personal computers, Apple does not allow the use of NFC security keys on Mac¹, therefore, the evaluation was conducted using a Windows device.

This allowed for evaluation under realistic usage conditions and comparison with results obtained from the external reader.

The evaluation considered not only the successful command processing, but also the correctness of the protocol, the latency of the response, and the overall success rate of communication exchanges. In addition, selected negative test cases involving unsupported or malformed APDU commands were performed to assess the robustness of the implementation. Particular attention was paid to timing constraints imposed by the ISO-DEP protocol, as these are critical for interoperability with mobile devices.

¹ <https://support.apple.com/en-us/102637>

5.2 Response Testing

The basic communication parameters of the emulated card were verified using the `nfc-poll` utility from the `libnfc` library. The tool was used with an external NFC reader (ACS ACR122U) to observe the activation sequence.

The following values were obtained during polling:

```
ATQA (SENS_RES): 00 44
UID (NFCID1):    5F 4C 4E 4B
SAK (SEL_RES):   20
ATS:             78 80 80 00
```

The **ATQA** value (0x0044) indicates a Type A target using a single-size UID (4-byte) and participation in the anti-collision and selection procedure. The reported UID consists of four bytes: 5F 4C 4E 4B, which correspond to the ASCII sequence “_LNK”. This static identifier was intentionally selected for testing and debugging purposes. In a production deployment, a unique identifier should be assigned to each hardware unit.

The **SAK** (0x20) signals that the device implements the ISO-DEP protocol.

The **ATS** sets the parameters of the upcoming transaction. The value of the lower nibble of the first byte, 0x78, represents FSCI (FSC Integer–8—means the FSC is 256 bytes). 0x80, as the value of the second byte, represents the bit rate; in this case, it is ISO-DEP_ATS_TA_SAME_D, meaning that the same bit rate is expected in both directions. The upper nibble of the third byte specifies 8 as WFI and the lower nibble 0 as SFGI.

Successful retrieval of these parameters demonstrates that the device correctly participates in the NFC-A activation sequence and exposes the protocol capabilities required for the operation of the Type 4 tag.

```
nfc-poll uses libnfc 1.8.0
NFC reader: ACS / ACR122U PICC Interface opened
NFC device will poll during 36000 ms (20 pollings of 300 ms for 6 modulations)
ISO/IEC 14443A (106 kbps) target:
  ATQA (SENS_RES): 00 44
  UID (NFCID1): 5f 4c 4e 4b
  SAK (SEL_RES): 20
  ATS: 78 80 80 00
Waiting for card removing...
```

Figure 5.1. Screenshot of the result of the `nfc-poll` command

5.3 Communication Capabilities Testing

5.3.1 CTAP Handling

The testing of CTAP functionality demonstrated stable and consistent behavior across multiple client platforms, including Apple iOS devices and Windows-based systems. The complete authentication workflow, comprising both credential registration (Make-Credential) and subsequent authentication (GetAssertion), was successfully executed in all evaluated scenarios. The correctness of the implementation was verified using platform-native WebAuthn interfaces and publicly available testing tools, ensuring that

```

CE: start TX successful
Received APDU (10 bytes):80100000000001040000
NFC: APDU parsed successfully
Handling extended APDU request: CLA=0x80 INS=0x10 P1=0x00 P2=0x00 Lc=1 Le=0
NFC: CTAP cmd=0x04 cbor_len=0
ctap_request: cmd=0x04 params_size=0
CTAP_CMD_GET_INFO
ctap_request: success in 6 ms, response length 176 bytes

```

Figure 5.2. Device log of processing an APDU containing a CTAP_GETINFO command.

the exchanged messages and responses adhered to the specifications defined by the FIDO Alliance.

For iOS and Windows devices, no deviations were observed from the expected behavior of the CTAP2 protocol, and the implementation exhibited complete interoperability with standard client platforms.

In contrast, limitations were identified when testing on Android-based devices. Experimental evaluation indicates that the interaction is limited to the legacy U2F protocol (Universal Second Factor). While CTAP2 functionality was successfully verified when the LionKey device was connected via USB, attempts to establish CTAP2-based communication over NFC consistently failed. Instead, the Android system, upon receiving a response associated with the U2F protocol, issued commands consistent with this protocol, suggesting a fallback to legacy behavior in accordance with earlier FIDO Alliance specifications. Since LionKey only supports the CTAP2 standard, it is impossible to comply with Android devices at this point.

These observations suggest that, as of April 2026, Android devices do not provide practical support for CTAP2 over NFC for external authenticators, effectively limiting NFC-based communication to U2F compatibility. Although Google introduced the full FIDO2 API for Android in 2025², this support primarily targets platform authenticators, and external authenticators connected via USB and NFC support remains vendor-specific. Since the LionKey implementation targets FIDO2 exclusively and does not implement backward compatibility with U2F, this platform constraint prevents successful interoperability with Android devices in the NFC use case.

5.3.1.1 Latency and Reliability

The response time was measured for the most computationally demanding CTAP command, CTAP_CMD_MAKE_CREDENTIAL. Across 50 consecutive trials, the average response time was 42 ms in the production mode and 270 ms in the debug mode, which is comfortably within the maximum reply time suggested by the FIDO Alliance of 800 ms. The significant increase in debug mode can be attributed primarily to logging overhead and reduced optimization levels, which introduce additional processing delays.

Regarding reliability, it was observed that the initiating device does not consistently detect premature termination of an NFC session. Specifically, when the authenticator is removed from the field of the initiator during an ongoing authentication procedure, the device correctly transitions to an idle state. However, the client application continues to wait for a response and does not recover from the interrupted communication.

As a consequence, subsequent authentication attempts cannot always be initiated, even

² <https://developers.google.com/identity/fido/android/native-apps>

though the user interface continues to indicate that NFC authentication is available. In this state, the NFC reader fails to re-establish communication with the authenticator, and recovery requires a full reset of the client context, typically by reloading the web page.

It was found that the frequency of this issue depends on the browser used. When tested with Safari, the failure occurred consistently after premature session termination. In contrast, Chrome and Firefox exhibited more robust behavior, requiring a page reload in approximately 40% of cases.

These observations suggest limited robustness in session termination and error handling within the client-side NFC stack. However, it cannot be conclusively determined to what extent this behavior is caused by the browser, the underlying operating system, or the implementation of the authenticator.

■ 5.3.2 NDEF Handling

The NDEF functionality was evaluated using a URI record to verify correct Type 4 Tag operation and interoperability with common NFC-enabled devices. The implementation was tested across multiple platforms, including iOS and Android devices, as well as an external NFC reader.

In all the scenarios tested, the NDEF message was successfully detected and processed, resulting in a 100% success rate in 50 of the tests conducted. The communication followed the standard Type 4 Tag access sequence defined by the NFC Forum, consisting of application selection, capability container access, and NDEF file retrieval.

More specifically, the observed APDU flow consisted of the following steps:

1. SELECT command for the NDEF application identifier
2. SELECT command for the Capability Container (CC) file
3. READ BINARY command to retrieve the CC file
4. SELECT command for the NDEF file
5. READ BINARY command to retrieve the NDEF message.

This sequence was executed consistently on all devices tested without deviation.

Response latency was measured at the APDU level in debug mode in 50 trials. The average response time was 25 ms for SELECT commands and 34 ms for READ BINARY commands. The increased latency for READ operations reflects the larger data payload and processing overhead associated with file retrieval. From the user perspective, no perceptible delay was observed during the interaction.

We also verified correct state machine behavior by manually sending APDUs containing both valid and invalid NDEF commands. This included checking proper handling of file selection and data access as well as handling of invalid APDUs.

■ 5.3.3 Use of CTAP on Real WebAuthn-enabled Websites

The implementation was further evaluated in real-world scenarios by it on WebAuthn-enabled websites. The authentication process was verified by authenticating on the WebAuthn demo and on Google.

```

CE: start waiting for command
CE: negotiated max frame size: 252
CE: start RX successful
Received APDU (13 bytes): 00A4040007D276000085010100
NFC: APDU parsed successfully
NFC: NDEF AID selected
Sent APDU (2 bytes): 9000
CE: start TX successful
Received APDU (7 bytes): 00A4000C02E103
NFC: APDU parsed successfully
NDEF: SELECT
Sent APDU (2 bytes): 9000
CE: start TX successful
Received APDU (5 bytes): 00B000000F
NFC: APDU parsed successfully
NDEF: READ
selected CC file
Response: offset=0 to_read=15 src=0 src_len=15 rsp_len=512
Sent APDU (17 bytes): 000F201000100004060001080000009000
CE: start TX successful
Received APDU (7 bytes): 00A4000C020001
NFC: APDU parsed successfully
NDEF: SELECT
Sent APDU (2 bytes): 9000
CE: start TX successful
Received APDU (5 bytes): 00B0000002
NFC: APDU parsed successfully
NDEF: READ
selected NDEF file
Response: offset=0 to_read=2 src=0 src_len=18 rsp_len=512
Sent APDU (4 bytes): 00109000
CE: start TX successful
Received APDU (5 bytes): 00B0000002
NFC: APDU parsed successfully
NDEF: READ
selected NDEF file
Response: offset=0 to_read=2 src=0 src_len=18 rsp_len=512
Sent APDU (4 bytes): 00109000
CE: start TX successful
Received APDU (5 bytes): 00B0000210
NFC: APDU parsed successfully
NDEF: READ
selected NDEF file
Response: offset=2 to_read=16 src=0 src_len=18 rsp_len=512
Sent APDU (18 bytes): D1010C55016C696F6E6865792E6465769000

```

Figure 5.3. Device log of NDEF APDU communication.

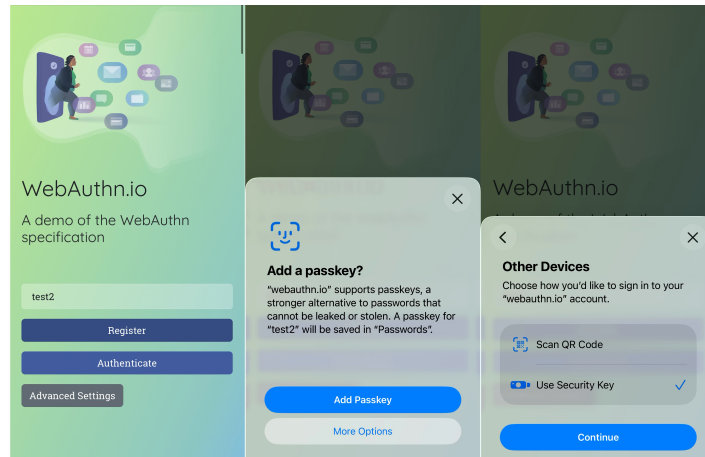


Figure 5.4. WebAuthn LionKey passkey registration flow via NFC on the iPhone.

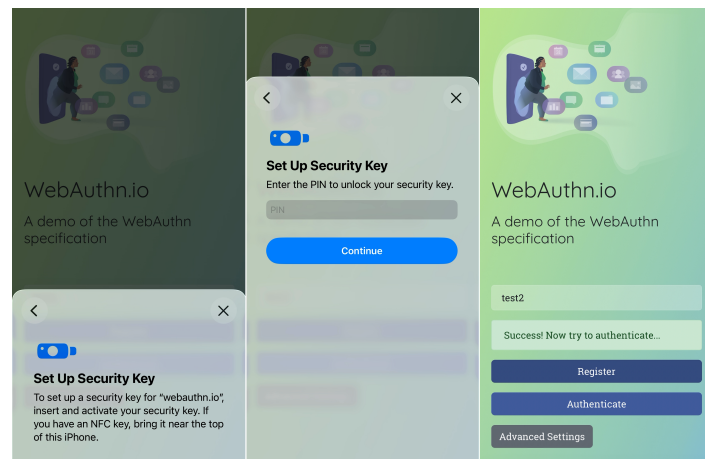


Figure 5.5. WebAuthn LionKey passkey registration flow via NFC on the iPhone.

Chapter 6

Conclusion

This thesis aimed to extend the LionKey hardware authenticator with Near Field Communication (NFC) capabilities, enabling its use in contactless authentication scenarios based on the FIDO2 CTAP standard. To achieve this, an implementation of NFC card emulation using ISO-DEP was designed and integrated into the existing system architecture.

The proposed solution was validated experimentally. As demonstrated in Chapter 5, the implementation supports both CTAP-based communication over NFC and NDEF-based data exchange. Correct functionality was confirmed at the protocol level by testing the APDU processing capabilities and at the application level through interoperability with multiple client platforms, including mobile devices and external NFC readers.

These results come with certain practical constraints worth noting. In particular, the reliability of NFC-based authentication is affected by inconsistent session termination handling on the client side, which may require manual recovery. Furthermore, it was discovered that not all Android devices support the CTAP2 protocol. In addition, the current hardware platform does not support energy harvesting in combination with the required communication capabilities, limiting the feasibility of a fully passive device and ruling out commercial viability in its current form.

Addressing these constraints fully would require a secure element-based platform that would integrate both NFC and cryptographic functionality. Such a transition would necessitate a complete redesign of the current system architecture and is therefore beyond the scope of this thesis.

In conclusion, the objectives of this thesis were met successfully. The presented implementation demonstrates that the integration of NFC-based communication into an existing hardware authenticator is feasible and provides a functional foundation for the further development of contactless security keys.

References

- [1] ITU. *Global and regional ICT data*. March, 2026.
https://www.itu.int/en/ITU-D/Statistics/Documents/facts/ITU_regional_global_Key_ICT_indicator_aggregates_Nov_2025.xlsx.
- [2] Verizon. *Data Breach Investigations Report*. 2025.
<https://www.verizon.com/business/resources/reports/2025-dbir-data-breach-investigations-report.pdf>.
- [3] Jumpcloud. *SME IT Trends Report*. 2024.
<https://jumpcloud.com/wp-content/uploads/2024/02/202401-EB-SMEITTrendsReport.pdf>.
- [4] Martin ENDLER. *FIDO2 USB bezpečnostní klíč*. 2025.
- [5] Sapio Research. *Consumer Password & Passkey Trends*. 2024.
<https://fidoalliance.org/wp-content/uploads/2024/05/World-Password-Day-2024-Report-FIDO-Alliance.pdf>.
- [6] 6sense . *Market Share of Yubico*. 2025.
<https://6sense.com/tech/two-factor-authentication/yubico-market-share>.
Archived at:
<https://perma.cc/MZ78-FJ62>.
- [7] Nina Bindel, Cas Cremers, and Mang Zhao. *FIDO2, CTAP 2.1, and WebAuthn 2: Provable Security and Post-Quantum Instantiation*. Cryptology ePrint Archive, Paper 2022/1029. 2022.
<https://eprint.iacr.org/2022/1029>.
- [8] Mouser Electronics. *RFID & NFC, An Introduction to Near Field Communications*. 2026.
https://eu.mouser.com/applications/rfid-nfc-introduction/?srsltid=AfmB0orUoW7dRVaiDpD7tnmaVleDhgMmboEuJqqAB8Rxx17e4JZ_PlCT.
- [9] Yubico. *YubiKey Technical Manual*. 2026.
<https://docs.yubico.com/hardware/yubikey/yk-tech-manual/webdocs.pdf>.
- [10] Google LLC. *Titan Security Key*. 2026.
<https://cloud.google.com/security/products/titan-security-key>.
- [11] Conrad Electronic. *All information about Near Field Communication via smart-phone & credit card*. November 12, 2024.
<https://www.conrad.com/en/guides/office-and-networking/communication-standards/nfc.html>.
- [12] Erik Hubers. *NFC Protocol Stack*. January 15, 2015.
https://commons.wikimedia.org/wiki/File:NFC_Protocol_Stack.png.
- [13] ISO/IEC. *ISO/IEC 14443-4:2018 Cards and security devices for personal identification — Contactless proximity objects — Part 4: Transmission protocol*. 2018.

- [14] FIDO Alliance. *Client to Authenticator Protocol (CTAP) Proposed Standard*. February 26, 2026.
<https://fidoalliance.org/specs/fido-v2.3-ps-20260226/fido-client-to-authenticator-protocol-v2.3-ps-20260226.html>.
- [15] STMicroelectronics. *STEVAL-25R3916B Data brief*. 2022.
https://www.st.com/resource/en/data_brief/steval-25r3916b.pdf.

Appendix A

Glossary

APDU	■ Application Protocol Data Unit
ASK	■ Amplitude Shift Keying
ATS	■ Answer To Select
BPSK	■ Binary Phase Shift Keying
CBOR	■ Concise Binary Object Representation
CE	■ Card Emulation
CID	■ Card Identifier
CLA	■ Class Byte
CTAP	■ Client to Authenticator Protocol
CTU FEE	■ Czech Technical University in Prague, Faculty of Electrical Engineering
DEP	■ Data Exchange Protocol
DID	■ Device Identifier
EM	■ Electromagnetic
FGT	■ Frame Guard Time
FIDO	■ Fast IDentity Online Alliance
FSC/FSD	■ Frame Size Card/Device
FWT	■ Frame Waiting Time
HAL	■ Hardware Abstraction Layer
HB	■ Historical Bytes
HID	■ USB Human Interface Devices
IDE	■ Integrated Development Environment
IRQ	■ Interrupt Request
I2C	■ Inter-Integrated Circuit
LLCP	■ Logical Link Control Protocol
MCU	■ Microcontroller Unit
MFA	■ Multi-factor Authentication
NFC	■ Near Field Communication
OOP	■ Object-Oriented Programming
OS	■ Operating System
PCB	■ Printed Circuit Board
PQ	■ Post-Quantum
RATS	■ Request Answer To Select
RF	■ Radio Frequency
RFAL	■ RF Abstraction Layer
RFID	■ Radio-Frequency Identification
SE	■ Secure Element
SPI	■ Serial Peripheral Interface
SW	■ Status Word
UI	■ User Interface
USB	■ Universal Serial Bus
UV	■ User Verification

U2F	■ Universal 2nd factor
VHBR	■ Very High Bit Rate
WTX	■ Waiting Time Extension
2FA	■ Second-factor Authentication